



CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE · INDIA

PROCTO

A SECURE AND SCALABLE ASSESSMENT PLATFORM

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Mission

Christ University is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment

Vision

Excellence and Service

Core Values

Faith in God | Moral Uprightness
Love of Fellow Beings | Social Responsibility
Pursuit of Excellence

Introduction

PROCTO is a secure and scalable web-based online assessment platform designed to address the challenges of conducting reliable remote examinations in modern education. The system integrates essential learning management features such as course creation, student enrollment, and exam scheduling with advanced online proctoring mechanisms to ensure academic integrity.

PROCTO enforces a controlled examination environment through full-screen mode and browser tab-switch restrictions while continuously monitoring webcam feeds and screen activity. Suspicious events such as camera interruptions or loss of exam window focus are automatically detected and logged with timestamps and visual evidence. Faculty members can monitor exams in real time through a centralized dashboard, evaluate descriptive answers manually, and rely on automated grading for objective questions.

Built on a cloud-based architecture using React and Next.js, Node.js and Python, and Supabase for database and authentication, PROCTO supports role-based access control for Admins, Faculty, and Students. The platform aims to reduce faculty workload, improve accountability, and provide a practical, scalable solution for secure online examinations.



The Problem

Challenges in Current Online Assessment Systems



Manual Supervision

Heavy reliance on manual invigilation and lack of structured monitoring



Poor Evidence Collection

Insufficient evidence collection for academic misconduct verification



Workflow Integration

Poor integration between classroom management and examination processes



Security Gaps

Existing systems lack distraction-free environments and controlled access

Introducing PROCTO

A Unified Platform for Secure Online Assessment

PROCTO combines classroom management, exam publishing, real-time proctoring, and automated grading into a single, scalable cloud platform—reducing faculty workload while ensuring academic integrity.



End-to-End Exam Management



Real-Time Proctoring



Automated Violation Detection



Evidence-Based Review

Secure Exam Environment

Enforcing a Distraction-Free Assessment Experience



Full-Screen Mode

Enforced full-screen examination interface



Browser Restrictions

Prevents tab switching and new window opening



Controlled Access

Students can only access assigned exams



Violation Warnings

Real-time alerts for suspicious behavior

Live Monitoring Dashboard

Centralized Faculty Control Center



Live Student Status

Real-time view of all active exam sessions



Alert Notifications

Instant alerts for suspicious activities



Violation Summaries

Comprehensive violation count & reports



Progress Tracking

Monitor exam completion & timing

User Roles & Responsibilities



Admin

- System management
- User control & permissions
- Exam rule configuration
- Platform oversight



Faculty

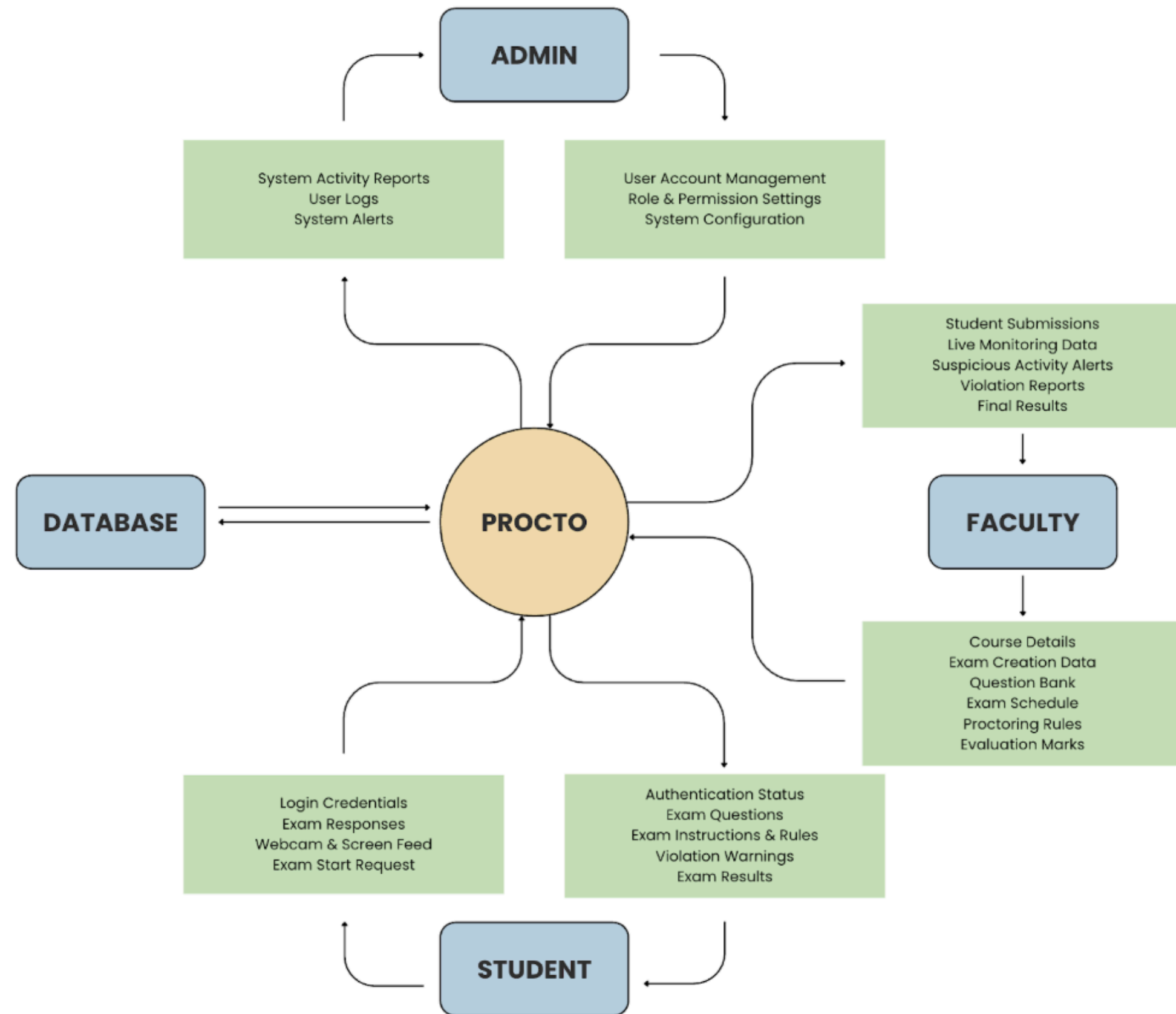
- Create & manage courses
- Schedule exams
- Monitor students live
- Evaluate & grade answers



Student

- Attend scheduled classes
- Take proctored exams
- Follow exam protocols
- View results & feedback

DFD 0



Technology Stack

Component	Technology
Platform	Web-based Cloud Application
Frontend	React - Next.js
Backend	Node.js, Python
Database	Supabase
Browsers	Chrome (recommended), Edge, Firefox
Server	Cloud-hosted with centralized storage

Key Benefits



Reduced Faculty Workload

Automated monitoring and grading streamline assessment process



Evidence-Based Review

Comprehensive logs with visual proof for fair evaluation



Secure & Reliable

Encrypted data transmission and secure authentication



Enhanced Academic Integrity

Real-time proctoring deters and detects misconduct



Scalable Architecture

Cloud infrastructure supports unlimited concurrent sessions



Comprehensive Analytics

Detailed insights into exam performance and violations

Exam Workflow

- 1 Exam Creation**
Faculty creates exam with MCQs and descriptive questions
- 3 Student Access**
Students enter secure full-screen exam mode
- 5 Auto-Save**
Periodic answer saving prevents data loss

- 2 Rule Configuration**
Define duration, timing, and proctoring rules
- 4 Live Monitoring**
Real-time proctoring with violation detection
- 6 Evaluation**
Auto-grading for MCQs, manual for descriptive

UI DESIGN



- Live exam monitoring in under 60 seconds

Next-gen proctoring for remote exams.

Procto lets faculty spin up secure, QR-enabled exam sessions and monitor activity in real time, while students join with a single code from any device.

Faculty login →

Student login

N

- QR-based session join
- Face & tab activity hooks
- Real-time integrity score

LIVE SESSION

Monitoring

SESSION CODE

PROCTO-4821

ACTIVE STUDENTS

32

JOIN IN 3 STEPS

1. Scan QR or enter code
2. Verify camera & mic
3. Start exam securely

Preview student view

ANOMALIES

3 flagged

FOCUS SCORE

92%

INTEGRITY

Stable

• High-Performance Exam Environment

Focus. Speed. Integrity.

Join your exam session instantly. No distractions, just performance. Your path to success starts here.

Enter Exam Lobby →

View Demo

STUDENT HUB

• Online



Join Classroom

Enter a class code to join



Join Quiz

Enter quiz code to start



Manage Classroom

View enrolled classes



• Designed for Educators & Administrators

Administer Secure Exams with Confidence.

Create exams in minutes and monitor integrity with AI-powered insights. Get real-time alerts, detailed reports, and complete control over your assessments.

Get Started Free →

Watch Demo

Faculty Dashboard

Manage your classrooms and assessments



Manage Classroom

View and edit your existing classrooms



Create new Classroom

Set up a new classroom for your course



Reports

View analytics and exam reports



P

Welcome to Procto

Sign in to access your exam session

Student

Faculty

Student Email

student@university.edu

Password

.....

☐ Remember me

[Forgot password?](#)

Enter Exam Lobby

or continue with

 Google

 GitHub

Don't have an account? [Sign up](#)

N

P

Welcome to Procto

Sign in to access your dashboard

Student

Faculty

Faculty Email

faculty@university.edu

Password

.....

☐ Remember me

[Forgot password?](#)

Access Dashboard

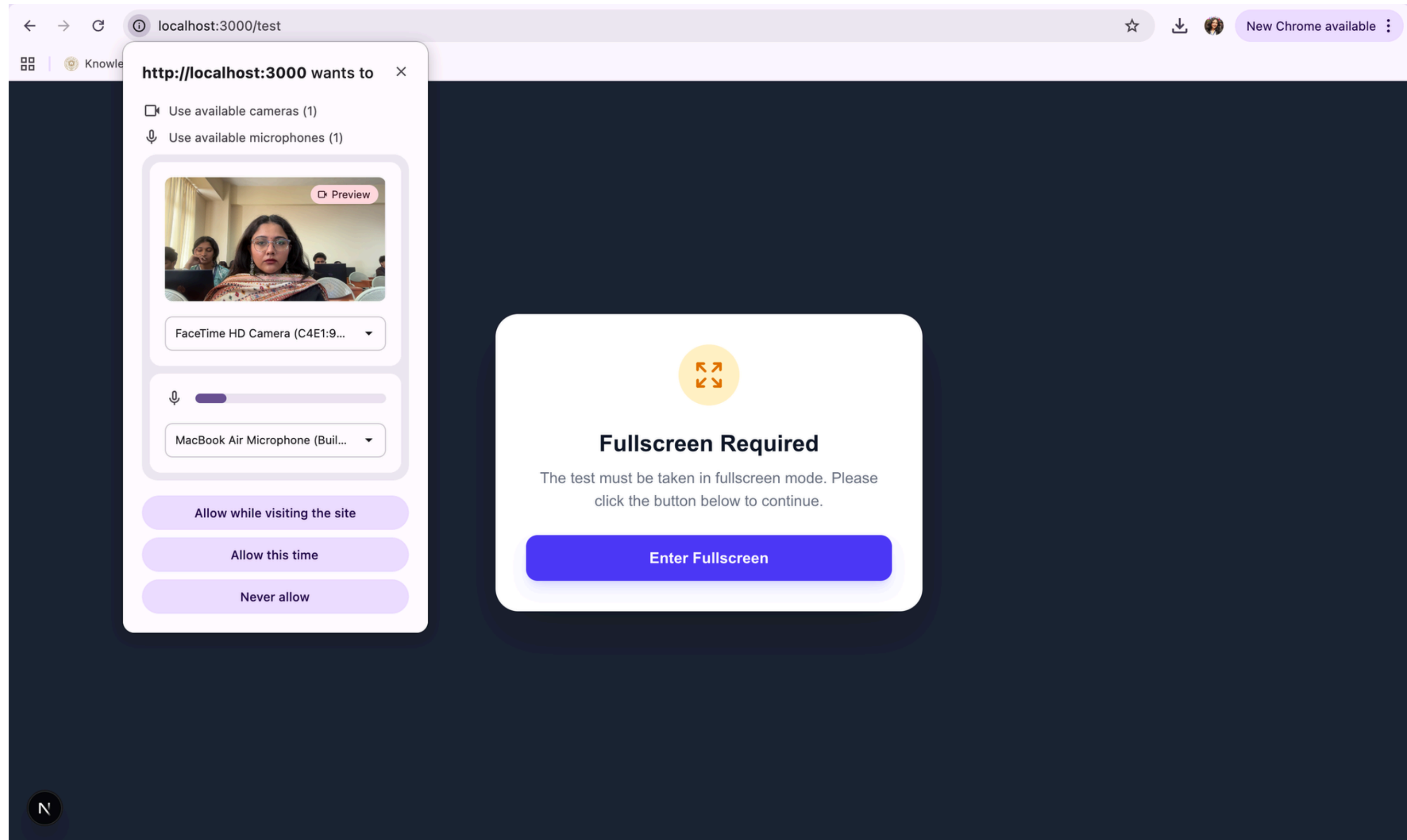
or continue with

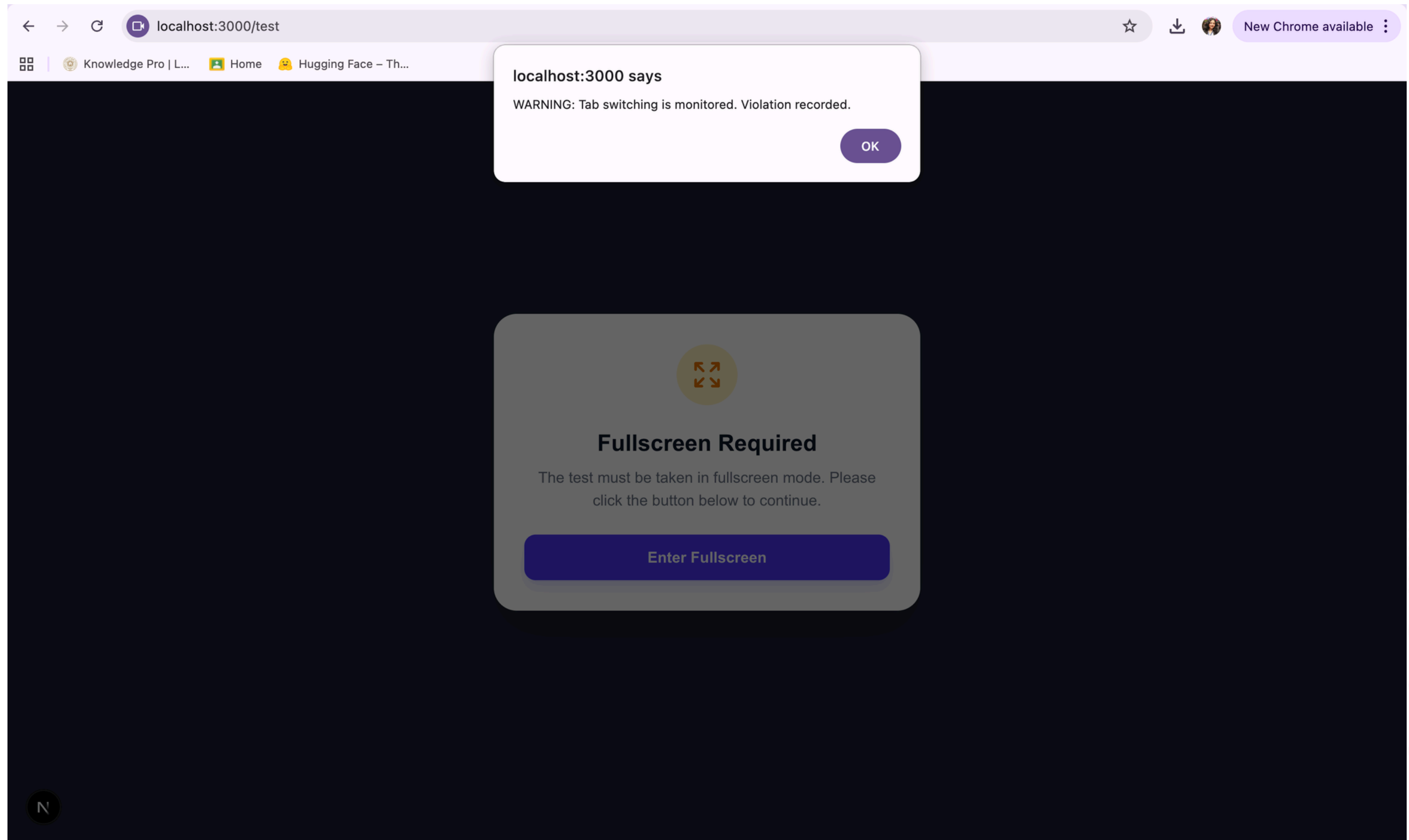
 Google

 GitHub

Don't have an account? [Sign up](#)

N





What is the time complexity of a binary search algorithm?

A $O(n)$

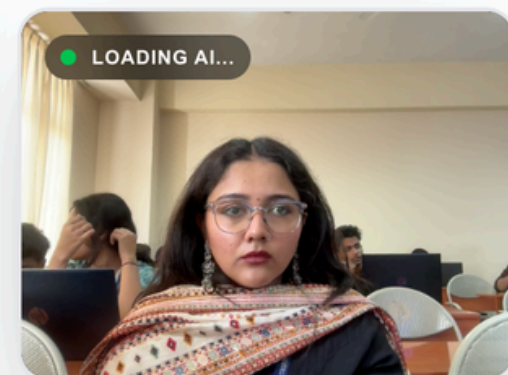
B $O(\log n)$



C $O(n^2)$

D $O(1)$

Next Question >



What is the time complexity of a binary search algorithm?

A $O(n)$

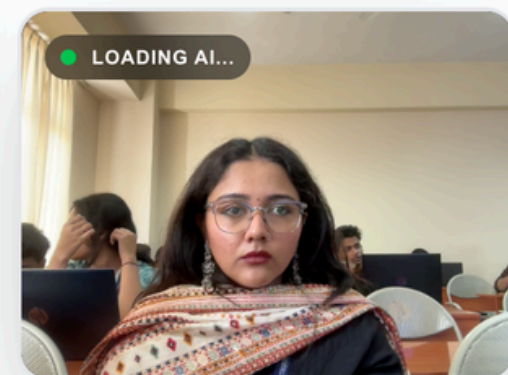
B $O(\log n)$



C $O(n^2)$

D $O(1)$

Next Question >



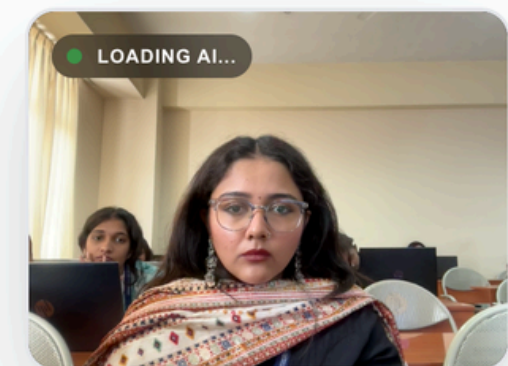
🛡️ Violations: 3



Test Submitted!

Your responses have been recorded. You may now close this window or wait for the proctor to dismiss you.

Return to Dashboard



Procto

FREE

Procto

main

PRODUCTION

Connect

Feedback

Search... K

Table Editor

schema public

+ New table

Search tables...

proctor_l... UNRESTRICTED

proctor_logs

Filter

Sort

Insert

RLS disabled

Index Advisor

Enable Realtime

Role postgres

	id uuid	event_type text	severity text	created_at timestamptz
	19012929-fb44-45b9-b1e8-59b6d25836d7	TAB_SWITCH	medium	2026-02-03 05:58:52.773428+00
	341e7414-6b69-4d1e-94eb-346c6f4a7cdf	TAB_SWITCH	medium	2026-02-03 05:59:25.331294+00
	3641f2cd-2c9b-4434-a509-b17b6e3a6e9c	NO_FACE	medium	2026-02-03 05:57:53.808603+00
	45fb4c91-1e88-4273-bbb9-046e364eaf6e	NO_FACE	medium	2026-02-03 05:59:05.767089+00
	5ecf55c7-8e9d-4165-b453-6aae20da1325	NO_FACE	medium	2026-02-03 05:59:15.372427+00
	617d08e8-785a-492c-bfcc-85c8db04e964	NO_FACE	medium	2026-02-03 05:58:05.759911+00
	71e7a0f9-218e-4365-92e6-29f174464163	EXIT_FULLSCREEN	medium	2026-02-03 05:59:23.470565+00
	7a1859e7-2e60-4c12-a774-36add9fca04	TAB_SWITCH	medium	2026-02-03 05:57:47.86193+00
	900dff88-f9a2-4037-81d1-ba16d8546afd	NO_FACE	medium	2026-02-03 05:58:17.812993+00
	a5dfa392-5cdf-4c7e-8e4d-6c55a27e8a89	TAB_SWITCH	medium	2026-02-03 05:59:18.81639+00
	cb112137-1f0a-440e-8307-4a4a4fa58b1e	NO_FACE	medium	2026-02-03 05:58:39.889805+00
	d598f0a5-82e3-4fb7-aa4b-d1f497c8b019	NO_FACE	medium	2026-02-03 05:58:51.323835+00
	d7cc68f0-efa7-4be0-a1dc-1cd2f96b0690	NO_FACE	medium	2026-02-03 05:58:29.743933+00

FUNCTIONAL REQUIREMENTS



User & Role Management

Course & Classroom Management

Exam Management

Secure Exam Environment

Proctoring & Monitoring

Live Monitoring Dashboard (Faculty)

Evaluation & Results

Reporting & Evidence Collection

NON-FUNCTIONAL REQUIREMENTS



Performance

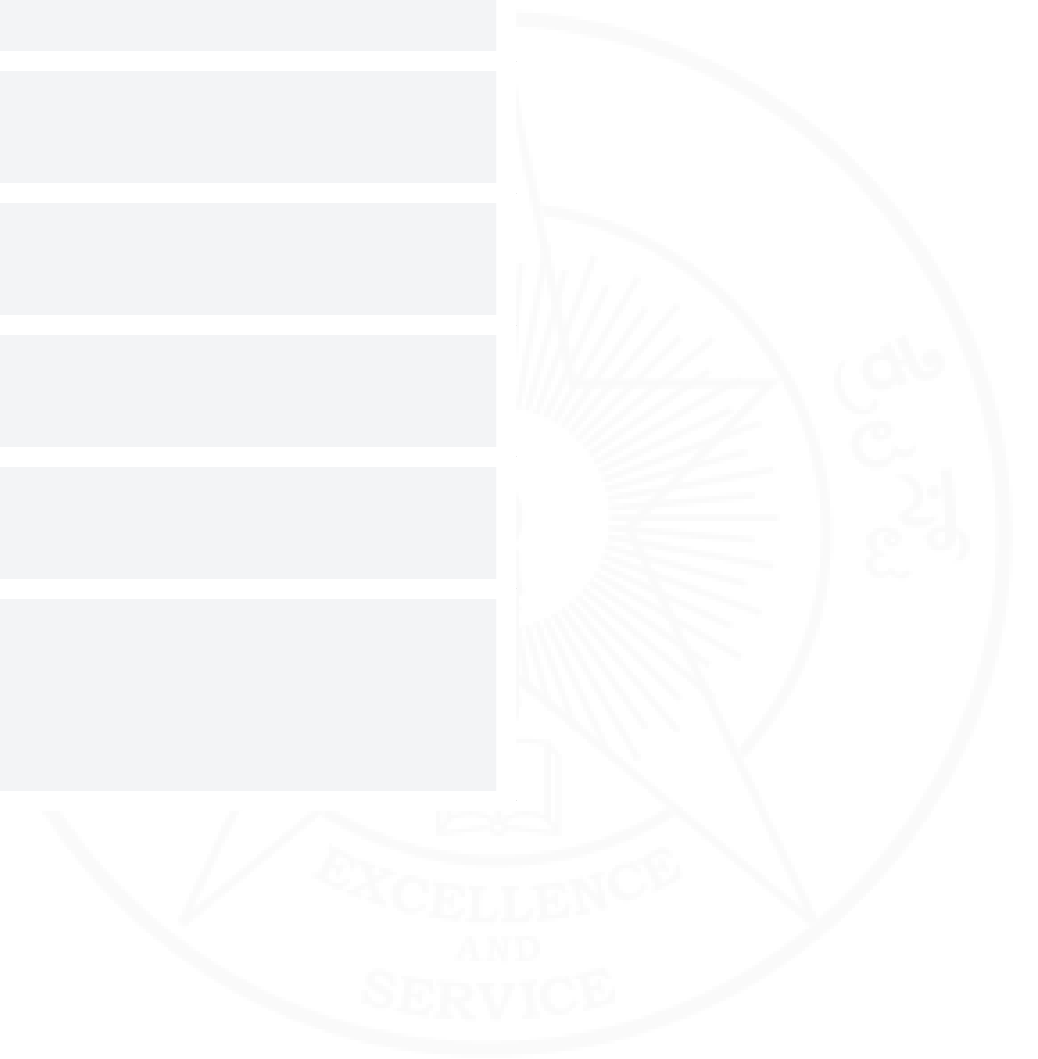
Security

Reliability

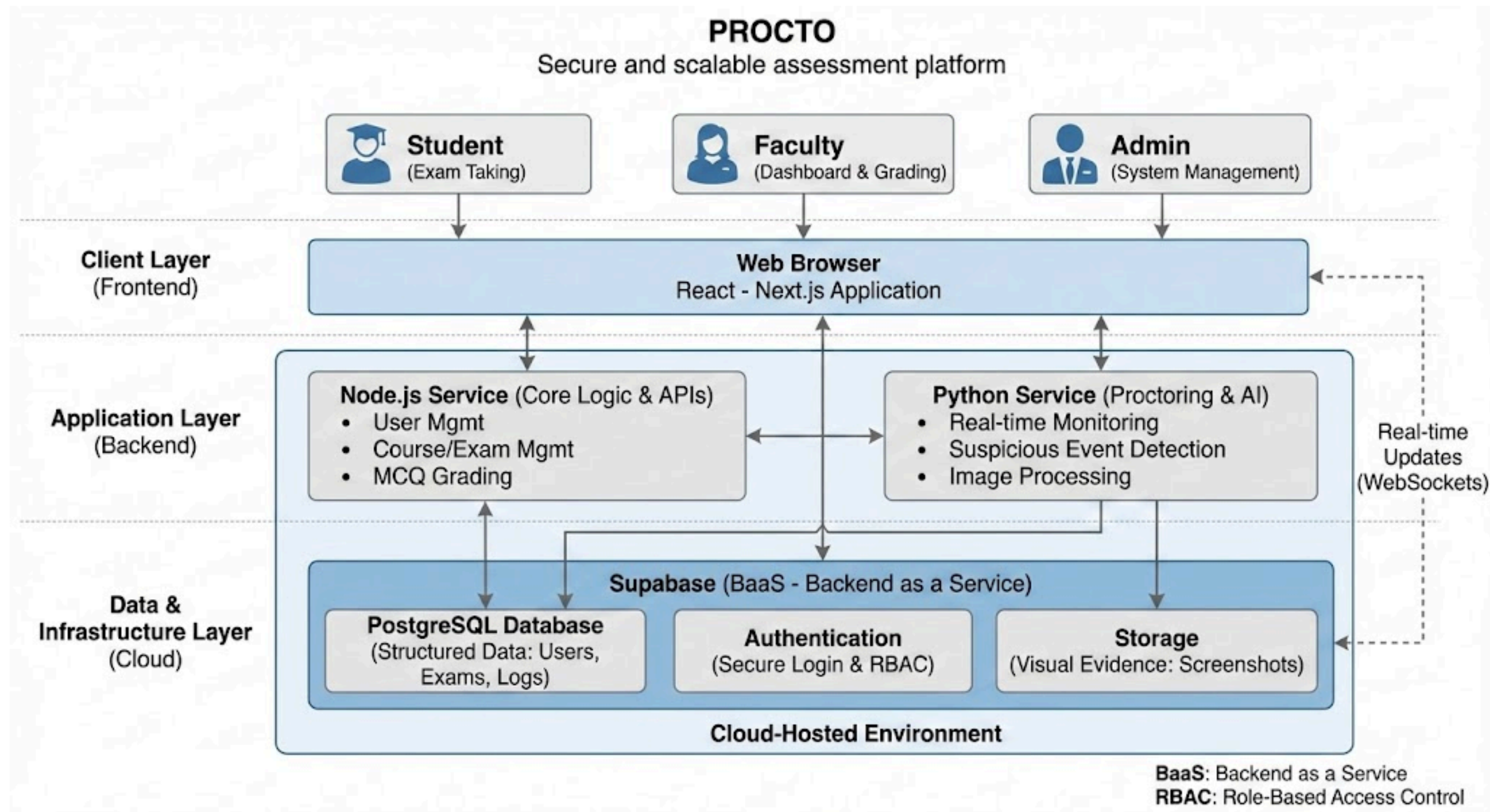
Usability

Scalability

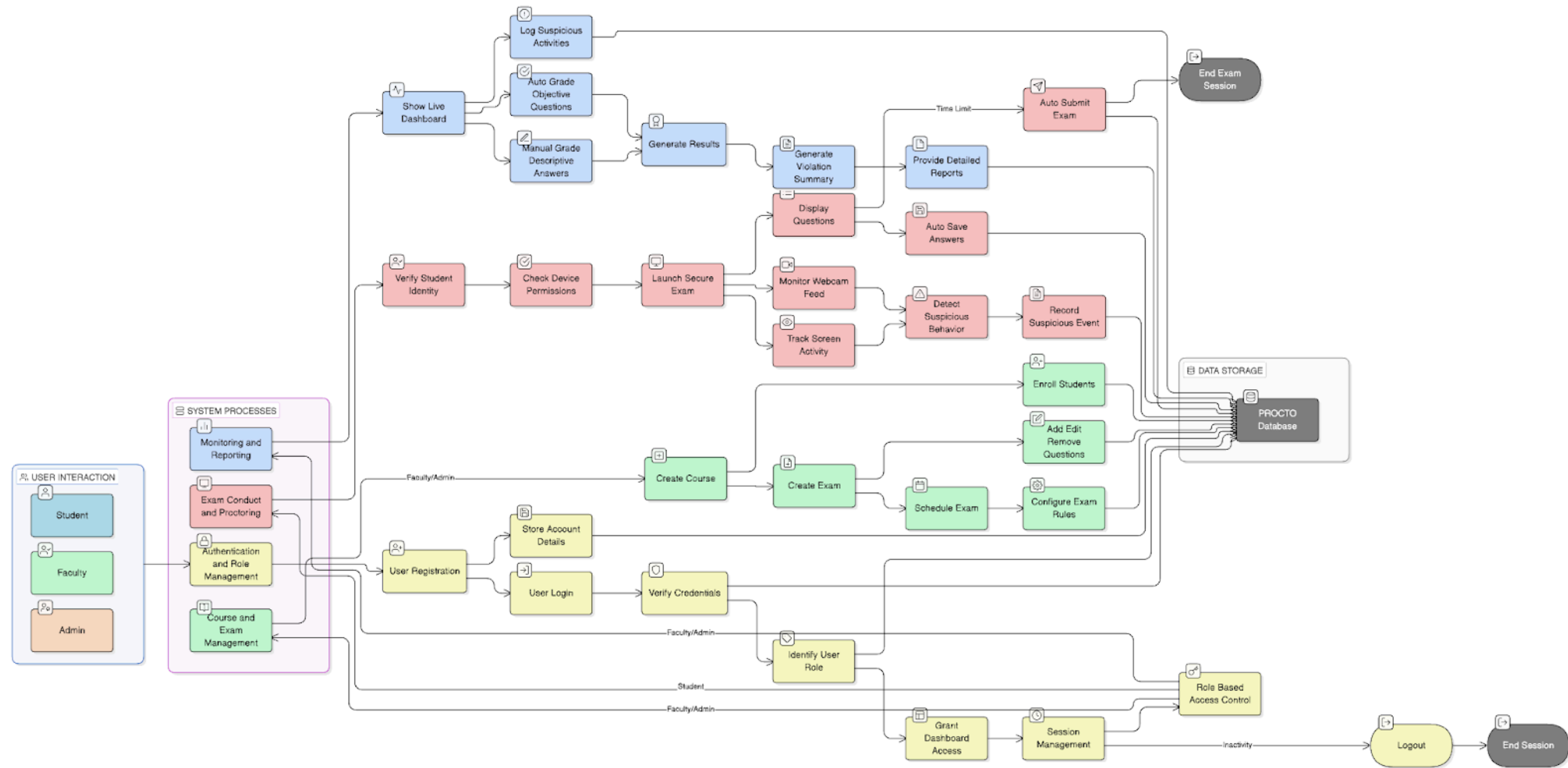
Maintainability



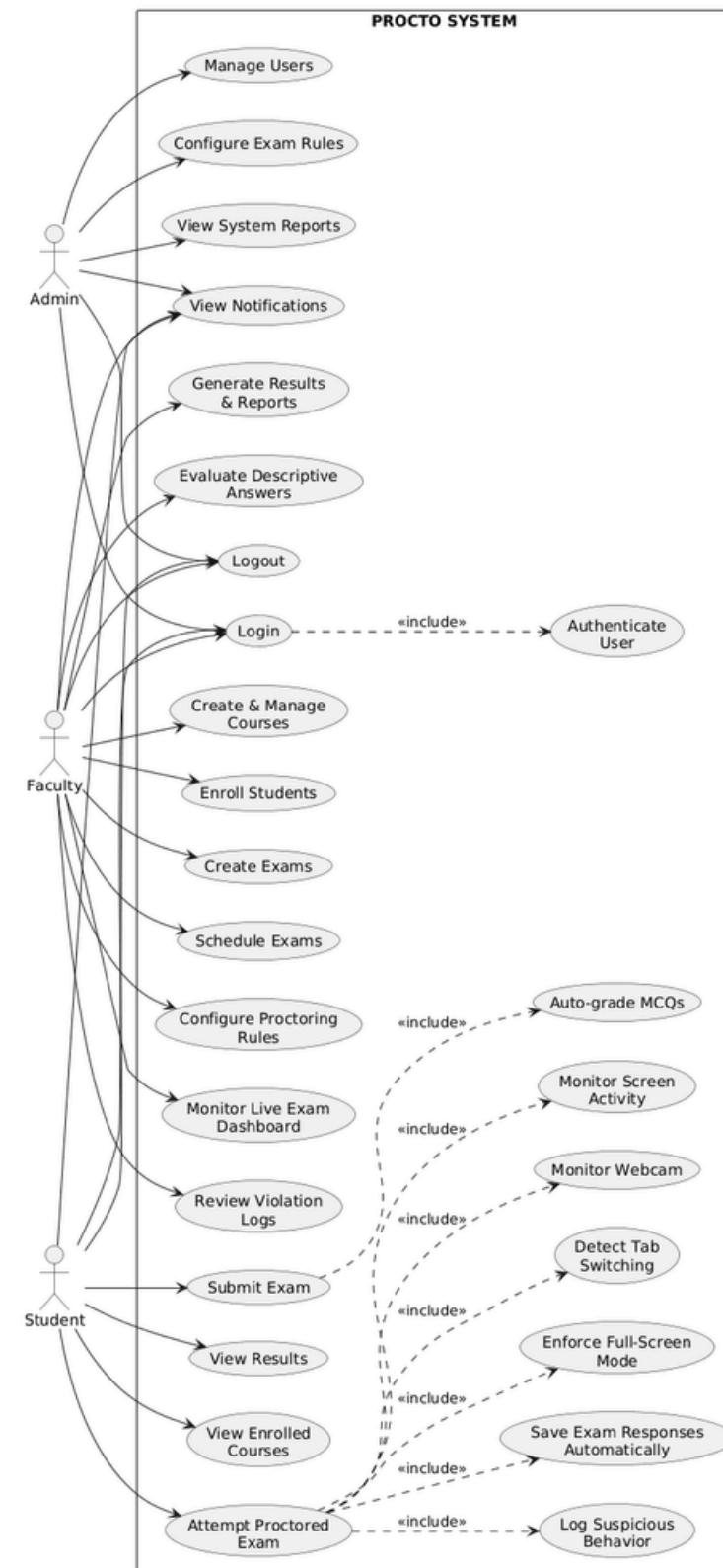
System Architecture Design



FLOW DIAGRAM



USE CASE DIAGRAM



Conclusion

The Software Requirement Specification for PROCTO defines a secure, scalable, and reliable online examination and proctoring system designed to uphold academic integrity in remote assessment environments. The system addresses key challenges of online examinations by integrating exam management, automated and live proctoring, evidence-based violation logging, and role-based access control within a unified platform.

By enforcing a controlled exam environment, enabling real-time monitoring, and supporting both automated and manual evaluation, PROCTO reduces faculty workload while ensuring transparency and accountability. The clearly defined functional and non-functional requirements provide a strong foundation for systematic development, testing, and future enhancement of the platform, making PROCTO a practical and effective solution for modern online education.



THANK YOU



